

CHANDIGARH UNIVERSITY

PROJECT REPORT

Spotify 2023 Data Analysis Dashboard using Power BI

A Data Visualization and Dashboard Project

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1. Introduction

Music streaming platforms record large amounts of data about songs, artists, playlists and listener activity. When this data is organised and visualised properly, it becomes easier to compare artists, study the performance of songs and understand the general trends in the music industry.

This project uses Microsoft Power BI to analyse Spotify 2023 song data. The analysis is presented as an interactive report of six pages. Each page is designed for a specific purpose: an overall summary, artist analysis, song analysis, platform analysis, analysis of music characteristics, and analysis by release year. The report uses cards, bar charts, column charts, a line chart, a pie chart, a scatter chart, a table and slicers to present the data.

This document records the design and purpose of each dashboard page. Screenshots of the actual Power BI pages are included, and the explanation given with each screenshot is limited to what is visible in it.

2. Project Objective

The objective of this project is to analyze Spotify 2023 song data using Microsoft Power BI and present meaningful insights through interactive dashboards, charts, cards, tables, filters, and visualizations.

To achieve this objective, the dashboard was organised around the following tasks:

- Summarising the dataset using key measures such as total streams, total songs, average streams, maximum streams and minimum streams.
- Comparing artists on the basis of total streams, average streams and number of songs.
- Examining individual songs by their total streams and by the number of entries for each track name.
- Studying the presence of songs in Spotify, Apple and Deezer playlists.
- Understanding the audio characteristics of songs, including danceability, energy, valence, acousticness, instrumentalness and speechiness.
- Observing how songs and streams are distributed across release years.

3. Dataset Description

The analysis is based on a Spotify 2023 song dataset. The Total Songs card on the Overview page shows that the dashboard covers 953 songs. The table below lists the fields that appear in the visuals of the dashboard, together with the pages on which they are used.

Field	Description	Used on
track_name	Name of the song	Song Analysis, Music Characteristics
artist(s)_name	Name of the artist or artists of the song	Overview, Artist Analysis, Platform Analysis, Music Characteristics
artist_count	Number of artists associated with a song (shown as legend values 1 to 8)	Release & Year Analysis
released_year	Year in which the song was released	Release & Year Analysis
streams	Number of streams of the song	Overview, Artist Analysis, Song Analysis, Release & Year Analysis
in_spotify_playlists	Playlist presence value for Spotify	Platform Analysis
in_apple_playlists	Playlist presence value for Apple	Platform Analysis
in_deezer_playlists	Playlist presence value for Deezer	Platform Analysis
danceability_%	Danceability of the song on a percentage scale	Platform Analysis, Music Characteristics
energy_%	Energy of the song on a percentage scale	Platform Analysis, Music Characteristics
valence_%	Valence of the song on a percentage scale	Music Characteristics
acousticness_%	Acousticness of the song on a percentage scale	Music Characteristics
instrumentalness_%	Instrumentalness of the song on a percentage scale	Music Characteristics
speechiness_%	Speechiness of the song on a percentage scale	Music Characteristics

Table 1: Dataset fields used in the dashboard

4. Data Cleaning and Preparation

Before the visualizations were created, the Spotify dataset was prepared and cleaned in Power Query. Power Query is the data transformation tool of Power BI and is used to load raw data, correct its structure and make it suitable for analysis.

The result of this process is a cleaned table named **spotify-2023-powerbi-ready**. This table was used as the data source for the Power BI analysis, and all measures and visuals described in this report are based on it.

Preparing the data in Power Query before building the report ensures that the visuals work on a consistent and structured table, and that the same prepared data is used for every page of the dashboard.

5. Power BI Measures

The following measures were used in the dashboard. They are created on the cleaned table and are displayed mainly as cards on the Overview page.

Measure	Definition	Value shown on Overview page
Total Streams	Sum of the streams column	489bn
Total Songs	Count of rows (songs) in the table	953
Average Streams	Average of the streams column	514.14M
Max Streams	Maximum value of the streams column	4bn
Min Streams	Minimum value of the streams column	3K

Table 2: Power BI measures used in the dashboard

The values in the third column are shown as displayed in the Overview screenshot, where large numbers are abbreviated by Power BI (for example, bn for billion, M for million and K for thousand).

6. Dashboard Overview

The Power BI report consists of six pages. The pages can be selected using the tabs at the bottom of the report. The table below gives a summary of the purpose of each page.

Page	Name	Main visuals	Purpose
1	Overview	Cards, bar chart, pie chart	Summary of key measures and top artists by total streams
2	Artist Analysis	Bar charts, column chart, slicer	Comparison of artists by streams, average streams and number of songs
3	Song Analysis	Bar chart, column chart, table, slicer	Analysis of individual songs by streams and number of entries
4	Platform Analysis	Column charts, scatter chart	Playlist presence across platforms and danceability versus energy
5	Music Characteristics	Cards, column chart	Audio characteristics of the songs
6	Release & Year Analysis	Column charts, line chart	Songs and streams across release years

Table 3: Summary of dashboard pages

Slicers are provided on the Artist Analysis and Song Analysis pages to filter the report by artist or by song. Each page is explained separately in the following sections.

7. Page 1 — Overview

The Overview page gives a summary of the complete dataset. It presents the main measures as cards and shows which artists have the highest total streams.

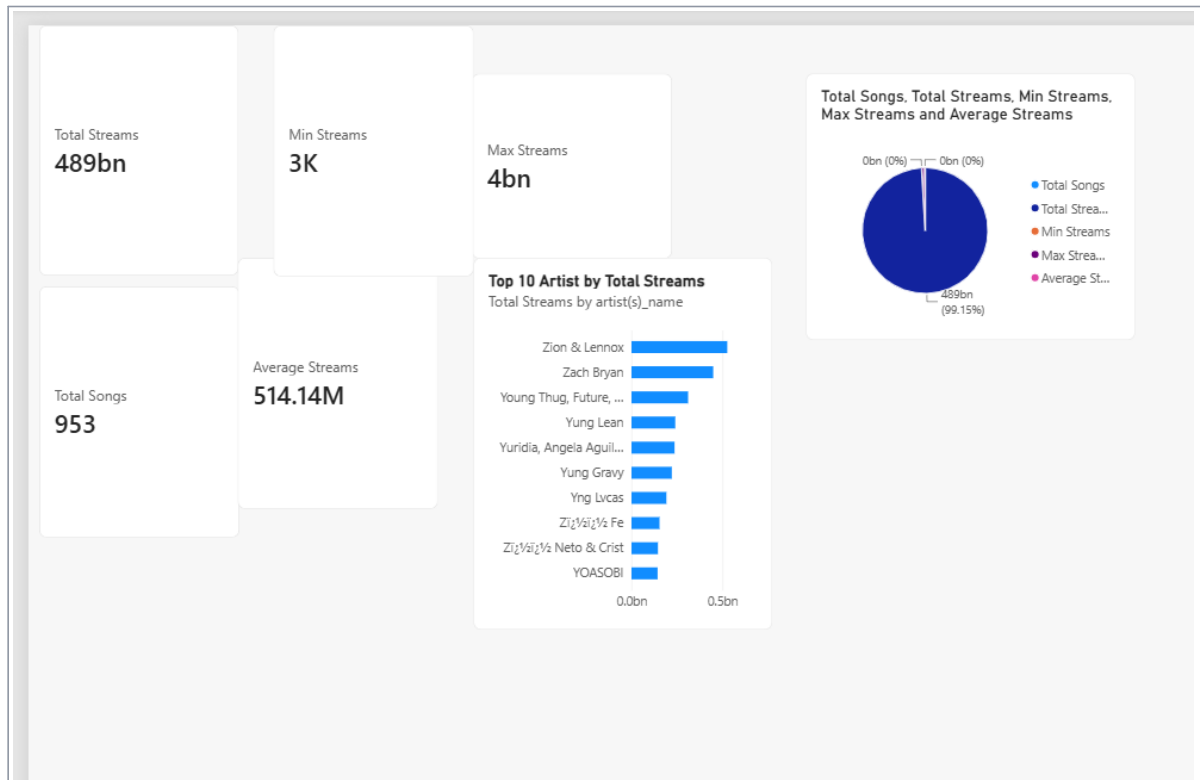


Figure 1: Overview Dashboard

7.1 Visuals on this Page

Visual	Measure / Field	Type of analysis
Total Streams card	Total Streams	Overall size of the dataset in streams
Total Songs card	Total Songs	Number of songs analysed
Average Streams card	Average Streams	Typical number of streams per song
Max Streams card	Max Streams	Highest streams value of a song
Min Streams card	Min Streams	Lowest streams value of a song
Top 10 Artist by Total Streams	Total Streams by artist(s)_name	Ranking of artists
Pie chart	Total Songs, Total Streams, Min Streams, Max Streams and Average Streams	Share of each measure in the combined total

7.2 Explanation

Total Streams. This card shows the sum of streams for all songs in the dataset. It is displayed as 489bn.

Total Songs. This card counts the songs in the table and shows 953. It indicates the number of records on which all other pages are based.

Average Streams. This card shows the average streams per song, displayed as 514.14M. It helps to judge the general level of streams for a song in the dataset.

Max Streams and Min Streams. These cards show the highest and the lowest streams value, displayed as 4bn and 3K respectively. Together they show the range between the most streamed and the least streamed song.

Top 10 Artist by Total Streams. This horizontal bar chart ranks artists by total streams. The artist names are shown on the vertical axis and the streams on the horizontal axis. It allows the top performing artists to be identified quickly.

Pie chart. This chart displays Total Songs, Total Streams, Min Streams, Max Streams and Average Streams together as shares of one pie.

7.3 Observations from the Screenshot

- The dashboard covers 953 songs with 489bn total streams and an average of 514.14M streams per song.
- The maximum streams value is displayed as 4bn and the minimum as 3K.
- In the Top 10 Artist by Total Streams chart, Zion & Lennox is at the top, followed by Zach Bryan and Young Thug, Future, Gunna.
- In the pie chart, Total Streams is labelled 489bn (99.15%), while the other measures are labelled 0bn (0%). This is because the measures have very different scales.

8. Page 2 — Artist Analysis

The Artist Analysis page is used to compare artists. It presents artist-wise streams, artist-wise number of songs and artists ranked by average streams, along with a slicer to filter by artist.

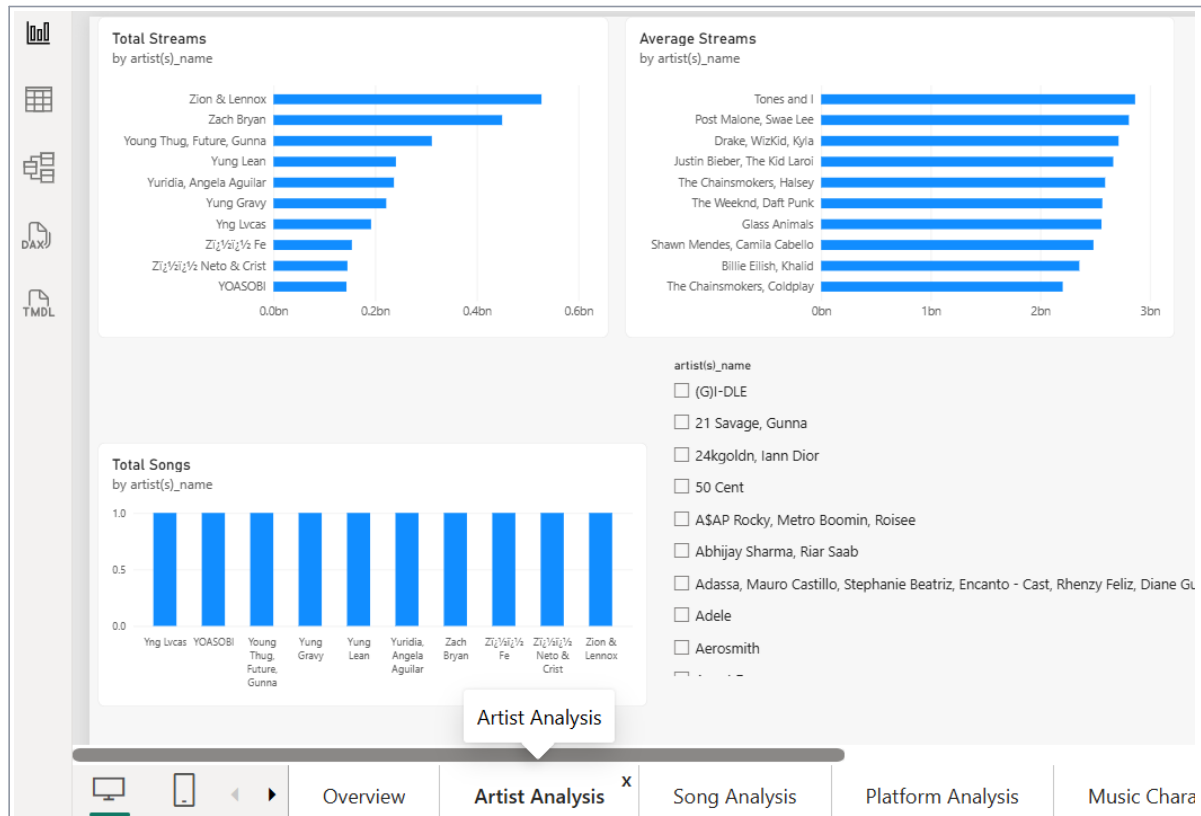


Figure 2: Artist Analysis Dashboard

8.1 Visuals on this Page

Visual	Measure / Field	Type of analysis
Total Streams by artist(s)_name	Total Streams	Artist-wise streams
Average Streams by artist(s)_name	Average Streams	Artists ranked by average streams
Total Songs by artist(s)_name	Total Songs	Artist-wise number of songs
Artist slicer	artist(s)_name	Filtering the page by artist

8.2 Explanation

Total Streams by artist(s)_name. This bar chart shows the ten artists with the highest total streams. It is used to compare how many streams each artist has accumulated.

Average Streams by artist(s)_name. This bar chart ranks artists by average streams. Average streams show the level of streams per song, so this chart gives a different view from total streams.

Total Songs by artist(s)_name. This column chart shows the number of songs for each artist. It gives the context for the streams charts, because the number of songs affects the total streams of an artist.

Artist filter. The slicer lists the values of artist(s)_name with check boxes. By default in Power BI, selecting one or more artists in a slicer filters the other visuals on the same page, so that a selected artist or group of artists can be examined separately.

Taken together, the three charts allow artists to be compared on streams, number of songs and average streams.

8.3 Observations from the Screenshot

- In Total Streams by artist(s)_name, Zion & Lennox has the longest bar at about 0.5bn, followed by Zach Bryan and Young Thug, Future, Gunna.
- In Average Streams by artist(s)_name, Tones and I has the longest bar at about 3bn, followed by Post Malone, Swae Lee and Drake, WizKid, Kyla. The ten bars shown lie roughly between 2bn and 3bn.
- In Total Songs by artist(s)_name, each of the ten artists shown has a value of 1.0.

9. Page 3 — Song Analysis

The Song Analysis page focuses on individual songs. It shows the most streamed songs, the number of entries for each track name, a table of songs and a slicer for selecting songs.

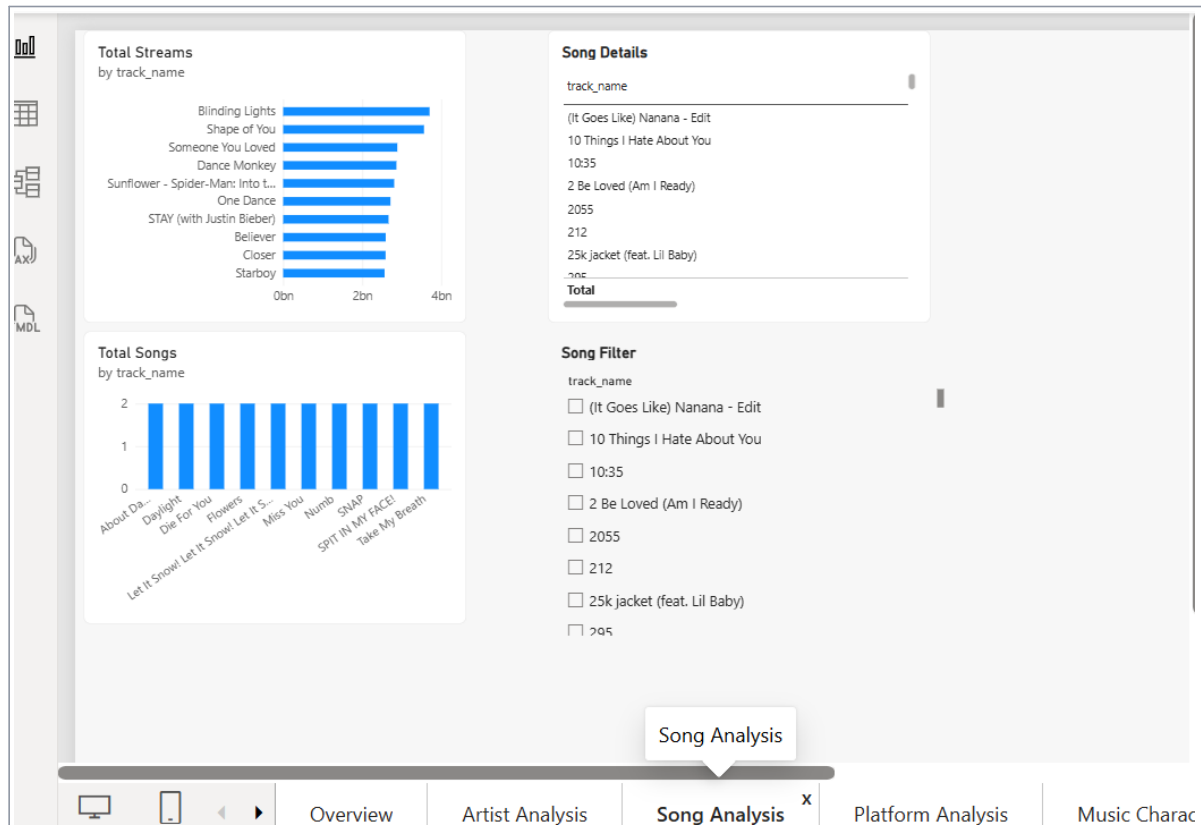


Figure 3: Song Analysis Dashboard

9.1 Visuals on this Page

Visual	Measure / Field	Type of analysis
Total Streams by track_name	Total Streams	Top 10 songs by total streams
Total Songs by track_name	Total Songs	Top 10 songs by number of entries
Song Details	track_name	Scrollable list of songs with a total row
Song Filter	track_name	Filtering the page by song

9.2 Explanation

Total Streams by track_name. This horizontal bar chart ranks songs by total streams. It is used to identify the songs with the highest streams and to compare them with each other.

Total Songs by track_name. This column chart shows the number of entries for each track name. It is used to see which track names appear more than once in the data.

Song Details. This table lists the songs by track_name in a scrollable form and includes a Total row at the bottom. It provides a simple way to browse the songs in the dataset.

Song Filter. The slicer lists the track names with check boxes. Selecting a song filters the other visuals on the page, so that an individual song can be analysed using its streams and its number of entries.

9.3 Observations from the Screenshot

- The songs shown in Total Streams by track_name are Blinding Lights, Shape of You, Someone You Loved, Dance Monkey, Sunflower - Spider-Man: Into t..., One Dance, STAY (with Justin Bieber), Believer, Closer and Starboy, in descending order of streams.
- Blinding Lights has the longest bar, and the axis extends up to 4bn.
- The Total Songs by track_name chart shows ten track names, each with a value of 2: About Da..., Daylight, Die For You, Flowers, Let It Snow! Let It S..., Miss You, Numb, SNAP, SPIT IN MY FACE! and Take My Breath.

10. Page 4 — Platform Analysis

The Platform Analysis page studies how songs are distributed across Spotify, Apple and Deezer playlists. It also compares danceability and energy for artists.

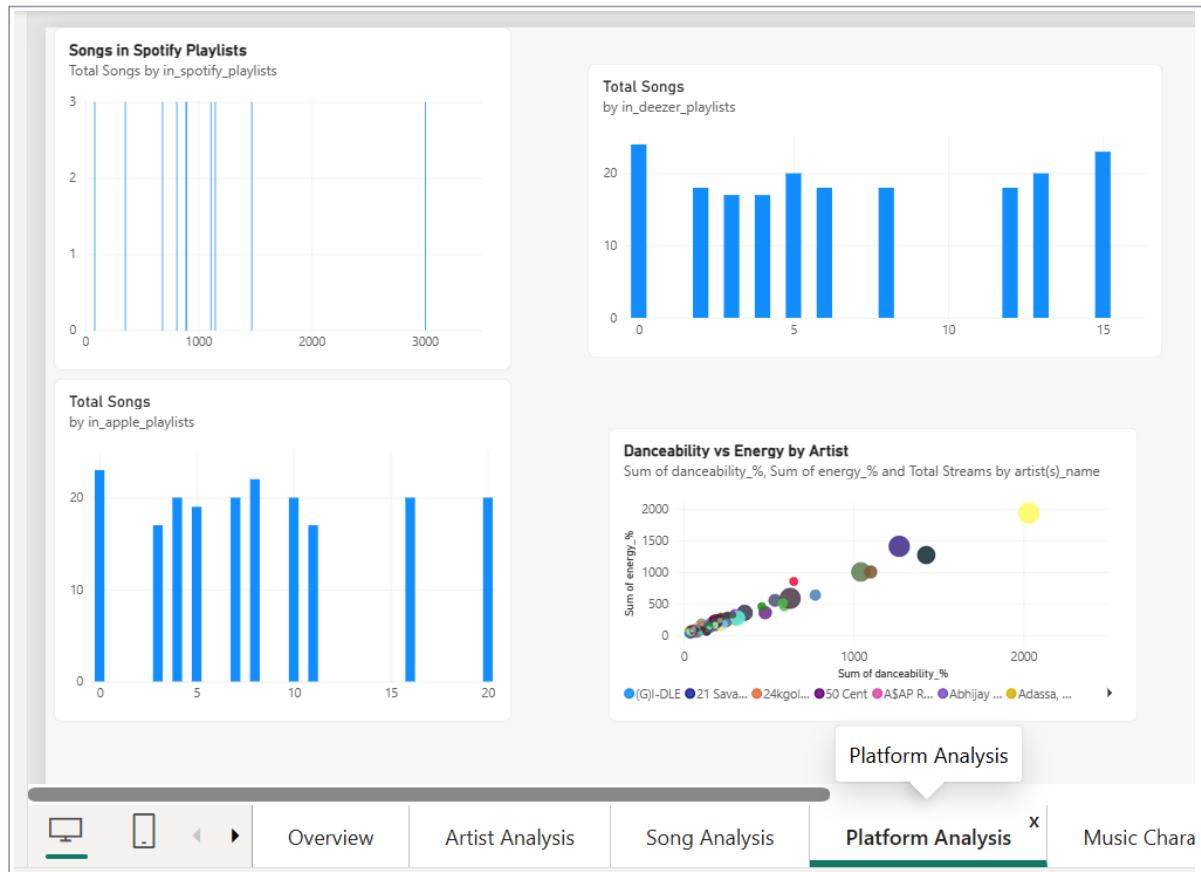


Figure 4: Platform Analysis Dashboard

10.1 Visuals on this Page

Visual	Measure / Field	Type of analysis
Songs in Spotify Playlists	Total Songs by in_spotify_playlists	Distribution of songs by Spotify playlist value
Total Songs by in_apple_playlists	Total Songs by in_apple_playlists	Distribution of songs by Apple playlist value
Total Songs by in_deezer_playlists	Total Songs by in_deezer_playlists	Distribution of songs by Deezer playlist value
Danceability vs Energy by Artist	Sum of danceability_%, Sum of energy_% and Total Streams by artist(s)_name	Relationship between danceability and energy

10.2 Explanation

Playlist presence across platforms. Three column charts show the number of songs (vertical axis) for each playlist value (horizontal axis) of the fields `in_spotify_playlists`, `in_apple_playlists` and `in_deezer_playlists`. Placing the three charts on one page makes it possible to compare the playlist presence of songs on the three platforms. The horizontal axes have different ranges on each platform, so the charts should be read separately for each platform.

Danceability vs Energy by Artist. This scatter chart plots the sum of `danceability_%` on the horizontal axis against the sum of `energy_%` on the vertical axis. Each point represents an artist (shown in the legend by `artist(s)_name`), and the size of the point represents Total Streams. The chart is used to see whether artists with higher danceability also have higher energy. Since the chart uses sums, the position of a point also depends on the number of songs of the artist.

10.3 Observations from the Screenshot

- In the Songs in Spotify Playlists chart the vertical axis ranges from 0 to 3 songs, and the horizontal axis extends beyond 3000. One bar appears near 3000.
- In the Apple chart the horizontal axis extends up to about 20, and in the Deezer chart up to about 15. The vertical axis in both charts reaches roughly 20 to 25 songs.
- In the scatter chart the points rise from the lower left to the upper right, which shows that higher danceability sums are generally accompanied by higher energy sums. Many points are placed close to the origin, and a few larger points are located towards the upper right.

11. Page 5 — Music Characteristics

The Music Characteristics page presents the audio characteristics of the songs as cards and lists the songs with the highest danceability.

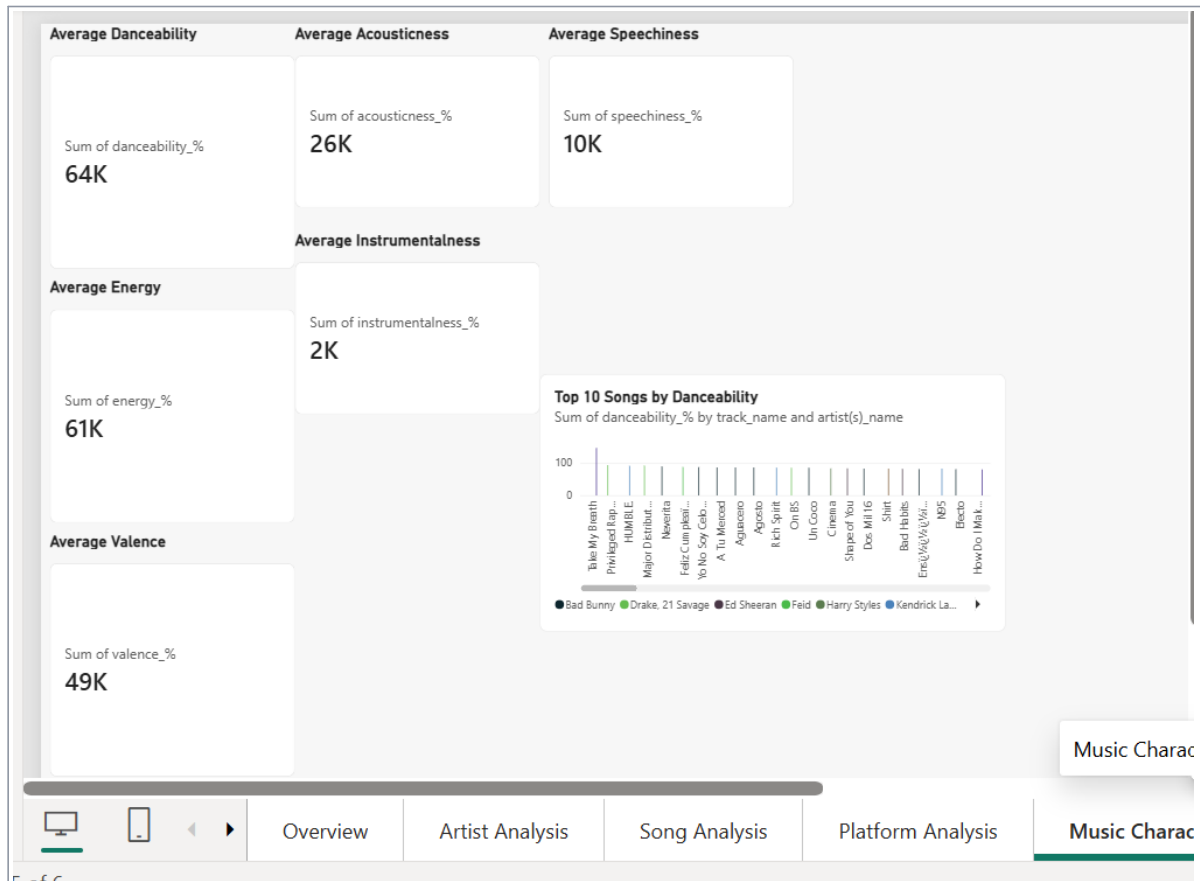


Figure 5: Music Characteristics Dashboard

11.1 Visuals on this Page

Visual (card title)	Field shown in card	Value displayed
Average Danceability	Sum of danceability_%	64K
Average Energy	Sum of energy_%	61K
Average Valence	Sum of valence_%	49K
Average Acousticness	Sum of acousticness_%	26K
Average Instrumentalness	Sum of instrumentalness_%	2K
Average Speechiness	Sum of speechiness_%	10K
Top 10 Songs by Danceability	Sum of danceability_% by track_name and artist(s)_name	Column chart

11.2 Explanation of the Audio Characteristics

- **Danceability** describes how suitable a song is for dancing.
- **Energy** describes the intensity and activity of a song.
- **Valence** describes the musical positiveness of a song, that is, whether it sounds cheerful or sad.
- **Acousticness** describes the extent to which a song is acoustic rather than electronic.
- **Instrumentalness** describes the likelihood that a song contains no vocals.
- **Speechiness** describes the presence of spoken words in a song.

These fields are stored on a percentage scale in the dataset. Viewing them together helps in understanding the musical properties of the songs, for example whether the songs in the dataset are mostly danceable and energetic, or mostly acoustic or instrumental.

The **Top 10 Songs by Danceability** column chart shows the danceability of songs by track_name and artist(s)_name. The legend at the bottom identifies artists by colour, and the chart can be scrolled horizontally.

11.3 Observations from the Screenshot

- Among the six cards, the displayed values are highest for danceability (64K), followed by energy (61K), valence (49K), acousticness (26K) and speechiness (10K). Instrumentalness is the lowest at 2K.
- The card titles read “Average”, while the field label inside each card reads “Sum of”. The values are reported here exactly as displayed.
- In the Top 10 Songs by Danceability chart, Take My Breath is the first song on the axis and has the tallest bar. Artists visible in the legend include Bad Bunny, Drake, 21 Savage, Ed Sheeran, Feid and Harry Styles.

12. Page 6 — Release & Year Analysis

The Release & Year Analysis page studies how songs and streams are distributed across release years, and how artist streams vary by year.

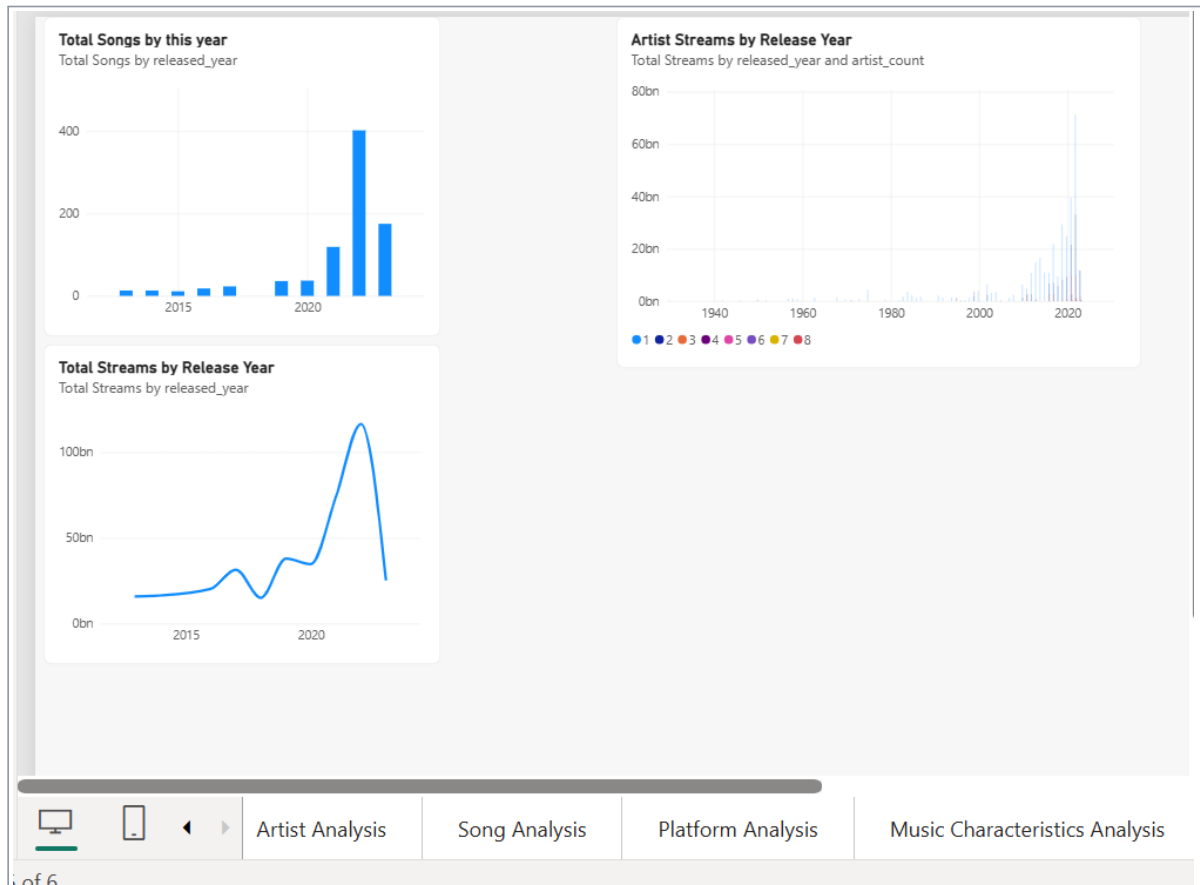


Figure 6: Release & Year Analysis Dashboard

12.1 Visuals on this Page

Visual	Measure / Field	Type of analysis
Total Songs by this year	Total Songs by released_year	Number of songs released in each year
Total Streams by Release Year	Total Streams by released_year	Trend of streams across release years
Artist Streams by Release Year	Total Streams by released_year and artist_count	Streams across years, split by artist count

12.2 Explanation

Total Songs by this year. This column chart shows the number of songs for each value of released_year. It indicates in which years most of the songs in the dataset were released.

Total Streams by Release Year. This line chart shows the total streams of the songs grouped by release year. A line chart is used because it makes the rise and fall of streams over time easy to follow.

Artist Streams by Release Year. This column chart plots total streams against released_year, and the legend separates the values by artist_count (1 to 8). It is used to see how streams are spread across years and across songs with a different number of artists.

12.3 Observations from the Screenshot

- In Total Songs by this year, the bars become higher towards the recent years. The tallest bar is close to 400 on the vertical axis, and the last bar is at about 175.
- In Total Streams by Release Year, the line rises to a peak of a little above 100bn, after which it falls sharply at the last point. The screenshot does not show whether the final year contains complete data, so this decline should be read with caution.
- In Artist Streams by Release Year, the horizontal axis starts from the 1940s, but most of the visible bars are concentrated in the years after 2000, and the highest bars reach about 70bn on the vertical axis.

13. Key Observations

The following observations are taken directly from the dashboard screenshots presented in this report.

13.1 Summary Measures

- The dataset contains 953 songs with a total of 489bn streams and an average of 514.14M streams per song.
- The maximum and minimum streams values are displayed as 4bn and 3K.

13.2 Artists

- Zion & Lennox, Zach Bryan and Young Thug, Future, Gunna are the top three artists by total streams.
- Artists ranked by average streams are led by Tones and I, with the ten bars shown lying roughly between 2bn and 3bn.
- The ten artists shown in the Total Songs by artist(s)_name chart each have one song.

13.3 Songs

- Blinding Lights and Shape of You are the two most streamed songs in the Song Analysis page.
- Ten track names, including Flowers, Numb and Take My Breath, appear twice in the Total Songs by track_name chart.

13.4 Platforms and Music Characteristics

- The Spotify, Apple and Deezer playlist charts use different horizontal ranges, so playlist values differ in scale across platforms.
- The Danceability vs Energy by Artist chart shows an upward relationship between the two characteristics.
- Among the six characteristic cards, danceability has the highest displayed value and instrumentality the lowest.

13.5 Release Years

- The number of songs and the streams are concentrated in the more recent release years.
- Total streams by release year show a peak followed by a sharp fall at the final point of the line.

13.6 Notes on Presentation

- Some artist names in the Artist Analysis and Overview pages are displayed with unreadable characters, for example “Ziç½iç½iç½ Fe”. This suggests a character encoding issue in the source data.

- The pie chart on the Overview page combines measures of very different scales, so Total Streams occupies almost the whole pie.
- The cards on the Music Characteristics page are titled “Average” but display “Sum of” the respective fields.

14. Conclusion

In this project, a six-page Power BI dashboard was developed for the Spotify 2023 dataset. The data was first prepared in Power Query and stored in the cleaned table spotify-2023-powerbi-ready. The measures Total Streams, Total Songs, Average Streams, Max Streams and Min Streams were then used together with different chart types to present the data.

The dashboard summarises the dataset on the Overview page and allows detailed comparison of artists and songs on the next two pages. The Platform Analysis page shows the distribution of songs across Spotify, Apple and Deezer playlists and compares danceability with energy. The Music Characteristics page presents the audio features of the songs, and the Release & Year Analysis page shows how songs and streams are distributed across release years.

The project shows how Power BI can be used to convert a raw music dataset into an interactive report that supports comparison and exploration of the data.

15. Learning Outcomes

Through this project, the following skills and concepts were learned:

- **Power BI dashboard creation:** Designing a multi-page report with cards, charts, tables and slicers, and arranging the visuals in a clear layout.
- **Data cleaning using Power Query:** Preparing a raw dataset before analysis and producing a clean table for reporting.
- **DAX measures:** Creating measures such as total, count, average, maximum and minimum to summarise the data.
- **Data visualization:** Selecting suitable visuals, such as bar charts, column charts, a line chart, a pie chart and a scatter chart, for different types of data.
- **Interactive filtering:** Using slicers to filter the report by artist and by song.
- **Comparative analysis:** Comparing artists, songs, platforms and release years using the same measures.
- **Understanding of Spotify music data:** Understanding fields such as streams, playlist presence and audio characteristics, and how they describe the performance and properties of songs.